Middle-Rung Law: physics and chemistry formulas written with the ether frequency v_e and the expansion rate H

Companion file to the OSF preregistration. Predictions fixed 2026-11-05. Author: John Lu Qianhe (约翰·路谦和). Normalization relation $4\pi^2r^3c^2/(T^2Ghv)=1$ after Zhang Xiangqian (2026, unpublished). All numbers use CODATA 2018 constants.

Middle-rung law. At every epoch t:
$\hbar \cdot G = c^5 / (4\pi^2 \cdot v_e(t) \cdot H(t))$, equivalently $v_e(t) \cdot H(t) = v_P^2$, with $v_P = \sqrt{(c^5/\hbar G)}/2\pi$ the Planck (human-boundary) frequency.
v_e and H both change with cosmic time; their product is fixed, so ħ and G stay constant.

0. Axioms and definitions (author, 2026-11-05/06)

1. **The universe consists only of ether structures and ether motion.** Different ether structures are what humans call different kinds of matter. Humans are ether structures too.
2. **Basic ether structure:** the regular tetrahedron. Every other structure (all matter) is a knot of it.
3. **Ether space around every structure — flow through the knot:** the basic ether is the finest unit ($v_e \approx 4.3\times10^{102}$ Hz, 10^{79} times the proton frequency), so it passes through every structure. It enters along the thumb axis, passes through the knot, and leaves spiralling outward at light speed in the plane of the four fingers, which curl counterclockwise around the thumb (right-hand rule); the pattern resembles dipole field lines. A knot is a persistent pattern in the flow, like a whirlpool, not a container, so no ether is created or destroyed (continuity holds), and every knot is carried by the flow with the same acceleration (equivalence principle). A macroscopic body contains vast numbers of knots with random axes, so its gravity is isotropic. Every structure, without exception, carries such a surrounding ether space.
4. **Relations between structures:** forces (gravitational, electromagnetic, …) are field-pressure differences where the ether spaces of different structures overlap. Flow contributions add in quadrature ($v^2 = 2\Phi$ superposes linearly); a uniform ether produces no force.
5. **Light speed is the background; only deviations are measurable.** The measurable part around a mass is the deviation flow $\sqrt{(2GM/r)}$ (11.2 km/s at Earth's surface). An outward flow of this speed has fluid acceleration $v\cdot dv/dr = -GM/r^2$, identical to an inward flow, so the outward-spiral reading reproduces the same exterior (Schwarzschild) geometry and all solar-system tests.
6. **Elements:** ether structure n = the element with atomic number n (1 = H … 118 = Og); every substance is "ether structure number + frequency", with frequency $f = mc^2/h$.
7. **Cosmic evolution of the ether frequency and the exact coasting transition 9.90 Gyr ago (at cosmic age 3.91 Gyr):** by the middle-rung law $v_e(t)\cdot H(t) = v_P^2 = 8.715\times10^{84}$ Hz², the ether frequency rises as the universe ages while the expansion rate falls, keeping their product invariant (and keeping ħ and G constant within the lunar laser ranging bound $|\dot{G}/G| < 1.5\times10^{-13}$

yr^{-1}). The evolution proceeds in two phases: (i) **Early non-uniform phase (from the Big Bang to cosmic age 3.912 Gyr [3.641–3.950 Gyr across H_0 = 61.8–63.5])**: prior to galaxy stabilization, the universe undergoes hot radiation/matter evolution and intense hierarchical merging of smaller structural units (within this phase, $v_e = 10^{96}$ Hz [8 orders × 12 dimensions] and 1.51×10^{96} Hz correspond to exact cosmic ages of **1,866 yr** and **2,833 yr**, or 3,636 yr and 5,491 yr under pure H = 1/t extrapolation). (ii) **Late uniform coasting phase (from 9.900 Gyr ago [9.893–9.933 Gyr ago] to the present)**: at cosmic age **3.912 Gyr** (z_s = 1.686, range 1.669–1.817; at the acoustic-angle benchmark z_s = 1.721, age 3.837 Gyr and lookback time 9.911 Gyr, where $1+z_s = 2.721 \approx e$), smaller structural units complete their assembly into stable large-scale galaxies, ether cancellation reaches its steady state ($N^{-2} \approx 2.2\times10^{-120}$, 10 orders × 12 dimensions), and expansion locks into uniform coasting $H(z) = H_0(1+z)$. Because the z_s-dependence in $t_0 - t_s = z_s/[H_0(1+z_s)]$ cancels to first order against $H_0 \propto \sqrt{(1+z_s)}$, the lookback time to stabilization is tightly pinned across the entire H_0 = 61.8–63.5 km s^{-1} Mpc^{-1} band to **9.893–9.933 Gyr ago (midpoint 9.900 Gyr ago)**. Over the subsequent 9.900 Gyr, $v_e(t)$ grows linearly with cosmic time from 1.615×10^{102} Hz to 4.337×10^{102} Hz while H(t) declines inversely, maintaining $v_e(t)\cdot H(t) = 8.715\times10^{84}$ Hz².

Open point: whether the thumb axis corresponds to spin/magnetic moment. Reading (identity, not a discovery): a knot frequency $f = mc^2/h = c/\lambda_C$, i.e. the number of light-speed ether passages per second across the knot's own Compton wavelength.

1. Present values (H_0 = 61.8–63.5 km s^{-1} Mpc^{-1}; central 62.0)

Quantity	Central (H_0 = 62.0)	Range (61.8–63.5)
Expansion rate H (universe rung)	2.009×10^{-18} s^{-1}	2.003×10^{-18} – 2.058×10^{-18} s^{-1}
Ether frequency $v_e = v_P^2/H$	4.337×10^{102} Hz	4.235×10^{102} – 4.351×10^{102} Hz
Planck (human-boundary) frequency v_P	2.952×10^{42} Hz	constant
Invariant product $v_e\cdot H = v_P^2$	8.715×10^{84} Hz²	constant
Rung ratio $N = v_P/H$	1.469×10^{60}	1.435×10^{60} – 1.474×10^{60}
Universe age (H_0t_0 = 0.876)	13.82 Gyr	13.49–13.86 Gyr
Mid-rung frequency $\sqrt{(H\ v_P)}$	2.435 THz	2.43–2.46 THz

Numerically: $\hbar\cdot G = 7.039\times10^{-45}$ J·m³ kg^{-1} s^{-2} = $c^5 / (4\pi^2 \cdot 4.337\times10^{102} \cdot 2.009\times10^{-18})$.

2. Physics formulas — symbolic form (v_e, H vary with time) and present-value form

Formula	Standard	Ether form (time-dependent $v_e(t)$, H(t))	Present-value form ($v_e\cdot H = 8.71\times10^{84}$ Hz²)	Value
Mass–energy	$E = mc^2$	$E = mc^2$ (unchanged); knot frequency $f = mc^2/h$	$E = mc^2$	8.988×10^{16} J (m = 1 kg)

Planck relation	$E = h\nu$	$E = c^5\nu / (2\pi G\, v_e\, H)$	$E = c^5\nu / (2\pi G \cdot 4.34\times10^{102} \cdot 2.01\times10^{-18})$	6.626×10^{-19} J ($\nu = 10^{15}$ Hz)
de Broglie wavelength	$\lambda = h/p$	$\lambda = c^5 / (2\pi G\, v_e\, H\, p)$	$\lambda = c^5 / (2\pi G \cdot 8.71\times10^{84} \cdot p)$	7.274×10^{-10} m (electron, 10^6 m/s)
Newtonian gravity	$F = GMm/r^2$	$F = c^5\, M\, m / (4\pi^2\, \hbar\, v_e\, H\, r^2)$	$F = c^5\, M\, m / (4\pi^2\, \hbar \cdot 8.71\times10^{84} \cdot r^2)$	9.82 N (1 kg at Earth surface)
Ether inflow = escape speed	$v = \sqrt{(2GM/r)}$	$v = \sqrt{(c^5 M / (2\pi^2\, \hbar\, v_e\, H\, r))}$	$v = \sqrt{(c^5 M / (2\pi^2\, \hbar \cdot 8.71\times10^{84} \cdot r))}$	11,190 m/s (Earth surface)
Gravitational redshift	$\Delta f/f = GM/(rc^2)$	$\Delta f/f = f_M\, f_r / (v_e\, H)$, $f_M = Mc^2/h$, $f_r = c/2\pi r$	$\Delta f/f = f_M\, f_r / 8.71\times10^{84}$	6.961×10^{-10} (Earth surface)
Schwarzschild radius	$r_s = 2GM/c^2$	$r_s = c^3 M / (2\pi^2\, \hbar\, v_e\, H)$	$r_s = c^3 M / (2\pi^2\, \hbar \cdot 8.71\times10^{84})$	2954 m (Sun)
Planck length	$\ell_P = \sqrt{(\hbar G/c^3)}$	$\ell_P = c / (2\pi \sqrt{(v_e\, H)})$	$\ell_P = c / (2\pi \sqrt{(8.71\times10^{84})})$	1.616×10^{-35} m
Planck time	$t_P = \sqrt{(\hbar G/c^5)}$	$t_P = 1 / (2\pi \sqrt{(v_e\, H)})$	$t_P = 1 / (2\pi \sqrt{(8.71\times10^{84})})$	5.391×10^{-44} s
Planck mass	$m_P = \sqrt{(\hbar c/G)}$	$m_P = 2\pi\, \hbar \sqrt{(v_e\, H)} / c^2$	$m_P = 2\pi\, \hbar \sqrt{(8.71\times10^{84})} / c^2$	2.176×10^{-8} kg
Black-hole entropy	$S = k_B c^3 A / (4\hbar G)$	$S = \pi^2\, k_B\, v_e\, H\, A / c^2$	$S = \pi^2\, k_B \cdot 8.71\times10^{84} \cdot A / c^2$	1.321×10^{46} J/K ($A = 1$ m^2)
Hawking temperature	$T = \hbar c^3 / (8\pi G M k_B)$	$T = \pi\, \hbar^2\, v_e\, H / (2c^2\, M\, k_B)$	$T = \pi\, \hbar^2 \cdot 8.71\times10^{84} / (2c^2\, M\, k_B)$	6.168×10^{-8} K (solar-mass black hole)
Fine-structure constant	$\alpha = e^2/(4\pi\varepsilon_0\hbar c)$	$\alpha = \pi\, e^2\, G\, v_e\, H / (\varepsilon_0\, c^6)$	$\alpha = \pi\, e^2\, G \cdot 8.71\times10^{84} / (\varepsilon_0\, c^6)$	7.297×10^{-3} (dimensionless; e = electron charge)
Bohr radius	$a_0 = 4\pi\varepsilon_0\hbar^2 / (m_e e^2)$	$a_0 = \varepsilon_0\, c^{10} / (4\pi^3\, G^2\, v_e^2\, H^2\, m_e\, e^2)$	$a_0 = \varepsilon_0\, c^{10} / (4\pi^3\, G^2\, (8.71\times10^{84})^2\, m_e\, e^2)$	5.292×10^{-11} m
Stefan–Boltzmann constant	$\sigma = \pi^2 k_B^4 / (60\hbar^3 c^2)$	$\sigma = \pi^2\, k_B^4\, (4\pi^2\, G\, v_e\, H)^3 / (60\, c^{17})$	$\sigma = \pi^2\, k_B^4\, (4\pi^2\, G \cdot 8.71\times10^{84})^3 / (60\, c^{17})$	5.670×10^{-8} W m^{-2} K^{-4}
Hubble law	$v = H\, d$	$v = H\, d$ (H = universe-rung frequency)	$v = 2.01\times10^{-18}$ s$^{-1} \cdot d$	62,000 m/s ($d = 1$ Mpc)
Normalization (Zhang 2026) on Hubble sphere	$4\pi^2 r^3 c^2 / (T^2 G h \nu) = 1$	$r = c/H$, $T = 2\pi/H \Rightarrow \nu = c^5/(hGH) = 2\pi\, v_e$	$\nu = 2\pi \cdot 4.34\times10^{102}$	2.725×10^{103} Hz
Temperature → frequency	$\nu = k_B T/h$	$\nu = 2\pi\, k_B\, T\, G\, v_e\, H / c^5$	$\nu = 2\pi\, k_B\, T\, G \cdot 8.71\times10^{84} / c^5$	6.459×10^{12} Hz ($T = 310$ K)
Two knots: gravity	$F = Gm_1m_2/r^2$	$F = c\, h\, f_1\, f_2 / (2\pi\, v_e\, H\, r^2)$	$F = c\, h\, f_1\, f_2 / (2\pi \cdot 8.71\times10^{84} \cdot r^2)$	6.674×10^{-11} N (two 1 kg, 1 m)

Electric / gravity ratio	$F_e/F_g = e^2/(4\pi\varepsilon_0 G m_1 m_2)$	$F_e/F_g = \alpha\, v_e\, H / (f_1\, f_2)$	$F_e/F_g = \alpha \cdot 8.71\times10^{84} / (f_1\, f_2)$	1.236×10^{36} (two protons)
Electric = gravity crossover	Stoney mass $\sqrt{\alpha}\, m_P$	$f = \sqrt{(\alpha\, v_e\, H)}$, $m = h\, f / c^2$	$f = \sqrt{(\alpha \cdot 8.71\times10^{84})}$	1.859×10^{-9} kg

Each ether form is an algebraic rewriting via the middle-rung law; the value column was checked to agree with the standard form to 10^{-9}. The physical content lies in the interpretation and in the dated predictions (P1–P5), not in the numerical agreement.

3. Chemistry formulas (natural constant e appears only in exponentials and logarithms)

Formula	Form
Ideal gas	$PV = nRT$, $R = N_A k_B$
Boltzmann distribution	$n_2/n_1 = e^{-h\Delta\nu/k_B T}$
Arrhenius	$k = A\, e^{-E_a/RT}$
Nernst	$E = E^0 - (RT/zF)\ln Q$; $2.303\, k_B T/e = 61.5$ mV per decade at 310 K
Gibbs / equilibrium	$\Delta G^0 = -RT \ln K$, $K = e^{-\Delta G^0/RT}$; $\Delta G = -nFE$
Proton-motive force	$\Delta p = \Delta\psi - (2.303RT/F)\,\Delta pH$
Radioactive decay	$N = N_0\, e^{-\lambda t}$, $t_{1/2} = \ln 2/\lambda$
Photon energy	$E = hc/\lambda = c^6 / (2\pi\, G\, v_e\, H\, \lambda)$
Rydberg constant	$R_\infty = \pi^3 m_e e^4 G^3 v_e^3 H^3 / (\varepsilon_0^2 c^{16})$
Bond energy → frequency	$\nu = \Delta E/h = 2\pi\, \Delta E\, G\, v_e\, H / c^5$

4. Elements as ether structures 1–118

Structure number = atomic number. Frequency $f = mc^2/h$. Ladder position $k = \log(m/m_U)/\log(m_P/m_U)$, with $m_U = hH/c^2$ and m_P the Planck mass. Ether form: $f = H^{1-k/2} \cdot v_e^{k/2}$ (exact). Values for $H_0 = 62.0$.

Z	Symbol	Atomic mass (u)	f (Hz)	k	Ether form
1	H	1.008	2.270×10^{23}	0.6823	$H^{0.6588}.v_e^{0.3412}$
2	He	4.0026	9.015×10^{23}	0.6923	$H^{0.6539}.v_e^{0.3461}$
3	Li	6.94	1.563×10^{24}	0.6962	$H^{0.6519}.v_e^{0.3481}$
4	Be	9.0122	2.030×10^{24}	0.6981	$H^{0.6509}.v_e^{0.3491}$
5	B	10.81	2.435×10^{24}	0.6994	$H^{0.6503}.v_e^{0.3497}$
6	C	12.011	2.705×10^{24}	0.7002	$H^{0.6499}.v_e^{0.3501}$
7	N	14.007	3.155×10^{24}	0.7013	$H^{0.6493}.v_e^{0.3507}$
8	O	15.999	3.604×10^{24}	0.7023	$H^{0.6489}.v_e^{0.3511}$

9	F	18.998	4.279×10^{24}	0.7035	$H^{0.6482}.\nu_e{}^{0.3518}$
10	Ne	20.18	4.545×10^{24}	0.7039	$H^{0.6480}.\nu_e{}^{0.3520}$
11	Na	22.99	5.178×10^{24}	0.7049	$H^{0.6476}.\nu_e{}^{0.3524}$
12	Mg	24.305	5.474×10^{24}	0.7053	$H^{0.6474}.\nu_e{}^{0.3526}$
13	Al	26.982	6.077×10^{24}	0.7060	$H^{0.6470}.\nu_e{}^{0.3530}$
14	Si	28.085	6.326×10^{24}	0.7063	$H^{0.6468}.\nu_e{}^{0.3532}$
15	P	30.974	6.976×10^{24}	0.7070	$H^{0.6465}.\nu_e{}^{0.3535}$
16	S	32.06	7.221×10^{24}	0.7073	$H^{0.6464}.\nu_e{}^{0.3536}$
17	Cl	35.45	7.985×10^{24}	0.7080	$H^{0.6460}.\nu_e{}^{0.3540}$
18	Ar	39.95	8.998×10^{24}	0.7089	$H^{0.6456}.\nu_e{}^{0.3544}$
19	K	39.098	8.806×10^{24}	0.7087	$H^{0.6456}.\nu_e{}^{0.3544}$
20	Ca	40.078	9.027×10^{24}	0.7089	$H^{0.6455}.\nu_e{}^{0.3545}$
21	Sc	44.956	1.013×10^{25}	0.7097	$H^{0.6451}.\nu_e{}^{0.3549}$
22	Ti	47.867	1.078×10^{25}	0.7102	$H^{0.6449}.\nu_e{}^{0.3551}$
23	V	50.942	1.147×10^{25}	0.7106	$H^{0.6447}.\nu_e{}^{0.3553}$
24	Cr	51.996	1.171×10^{25}	0.7108	$H^{0.6446}.\nu_e{}^{0.3554}$
25	Mn	54.938	1.237×10^{25}	0.7112	$H^{0.6444}.\nu_e{}^{0.3556}$
26	Fe	55.845	1.258×10^{25}	0.7113	$H^{0.6444}.\nu_e{}^{0.3556}$
27	Co	58.933	1.327×10^{25}	0.7117	$H^{0.6442}.\nu_e{}^{0.3558}$
28	Ni	58.693	1.322×10^{25}	0.7117	$H^{0.6442}.\nu_e{}^{0.3558}$
29	Cu	63.546	1.431×10^{25}	0.7122	$H^{0.6439}.\nu_e{}^{0.3561}$
30	Zn	65.38	1.473×10^{25}	0.7124	$H^{0.6438}.\nu_e{}^{0.3562}$
31	Ga	69.723	1.570×10^{25}	0.7129	$H^{0.6436}.\nu_e{}^{0.3564}$
32	Ge	72.63	1.636×10^{25}	0.7132	$H^{0.6434}.\nu_e{}^{0.3566}$
33	As	74.922	1.688×10^{25}	0.7134	$H^{0.6433}.\nu_e{}^{0.3567}$
34	Se	78.971	1.779×10^{25}	0.7138	$H^{0.6431}.\nu_e{}^{0.3569}$
35	Br	79.904	1.800×10^{25}	0.7139	$H^{0.6431}.\nu_e{}^{0.3569}$
36	Kr	83.798	1.887×10^{25}	0.7142	$H^{0.6429}.\nu_e{}^{0.3571}$
37	Rb	85.468	1.925×10^{25}	0.7144	$H^{0.6428}.\nu_e{}^{0.3572}$
38	Sr	87.62	1.974×10^{25}	0.7145	$H^{0.6427}.\nu_e{}^{0.3573}$

39	Y	88.906	2.002×10^{25}	0.7147	$H^{0.6427}.\nu_e{}^{0.3573}$
40	Zr	91.224	2.055×10^{25}	0.7148	$H^{0.6426}.\nu_e{}^{0.3574}$
41	Nb	92.906	2.093×10^{25}	0.7150	$H^{0.6425}.\nu_e{}^{0.3575}$
42	Mo	95.95	2.161×10^{25}	0.7152	$H^{0.6424}.\nu_e{}^{0.3576}$
43	Tc	98	2.207×10^{25}	0.7154	$H^{0.6423}.\nu_e{}^{0.3577}$
44	Ru	101.07	2.276×10^{25}	0.7156	$H^{0.6422}.\nu_e{}^{0.3578}$
45	Rh	102.91	2.318×10^{25}	0.7157	$H^{0.6421}.\nu_e{}^{0.3579}$
46	Pd	106.42	2.397×10^{25}	0.7159	$H^{0.6420}.\nu_e{}^{0.3580}$
47	Ag	107.87	2.430×10^{25}	0.7160	$H^{0.6420}.\nu_e{}^{0.3580}$
48	Cd	112.41	2.532×10^{25}	0.7163	$H^{0.6418}.\nu_e{}^{0.3582}$
49	In	114.82	2.586×10^{25}	0.7165	$H^{0.6418}.\nu_e{}^{0.3582}$
50	Sn	118.71	2.674×10^{25}	0.7167	$H^{0.6416}.\nu_e{}^{0.3584}$
51	Sb	121.76	2.742×10^{25}	0.7169	$H^{0.6415}.\nu_e{}^{0.3585}$
52	Te	127.6	2.874×10^{25}	0.7173	$H^{0.6414}.\nu_e{}^{0.3586}$
53	I	126.9	2.858×10^{25}	0.7172	$H^{0.6414}.\nu_e{}^{0.3586}$
54	Xe	131.29	2.957×10^{25}	0.7175	$H^{0.6413}.\nu_e{}^{0.3587}$
55	Cs	132.91	2.994×10^{25}	0.7176	$H^{0.6412}.\nu_e{}^{0.3588}$
56	Ba	137.33	3.093×10^{25}	0.7178	$H^{0.6411}.\nu_e{}^{0.3589}$
57	La	138.91	3.129×10^{25}	0.7179	$H^{0.6411}.\nu_e{}^{0.3589}$
58	Ce	140.12	3.156×10^{25}	0.7179	$H^{0.6410}.\nu_e{}^{0.3590}$
59	Pr	140.91	3.174×10^{25}	0.7180	$H^{0.6410}.\nu_e{}^{0.3590}$
60	Nd	144.24	3.249×10^{25}	0.7181	$H^{0.6409}.\nu_e{}^{0.3591}$
61	Pm	145	3.266×10^{25}	0.7182	$H^{0.6409}.\nu_e{}^{0.3591}$
62	Sm	150.36	3.387×10^{25}	0.7184	$H^{0.6408}.\nu_e{}^{0.3592}$
63	Eu	151.96	3.423×10^{25}	0.7185	$H^{0.6407}.\nu_e{}^{0.3593}$
64	Gd	157.25	3.542×10^{25}	0.7188	$H^{0.6406}.\nu_e{}^{0.3594}$
65	Tb	158.93	3.580×10^{25}	0.7188	$H^{0.6406}.\nu_e{}^{0.3594}$
66	Dy	162.5	3.660×10^{25}	0.7190	$H^{0.6405}.\nu_e{}^{0.3595}$
67	Ho	164.93	3.715×10^{25}	0.7191	$H^{0.6404}.\nu_e{}^{0.3596}$
68	Er	167.26	3.767×10^{25}	0.7192	$H^{0.6404}.\nu_e{}^{0.3596}$

69	Tm	168.93	3.805×10^{25}	0.7193	$H^{0.6404}.v_e^{0.3596}$
70	Yb	173.05	3.898×10^{25}	0.7195	$H^{0.6403}.v_e^{0.3597}$
71	Lu	174.97	3.941×10^{25}	0.7195	$H^{0.6402}.v_e^{0.3598}$
72	Hf	178.49	4.020×10^{25}	0.7197	$H^{0.6402}.v_e^{0.3598}$
73	Ta	180.95	4.076×10^{25}	0.7198	$H^{0.6401}.v_e^{0.3599}$
74	W	183.84	4.141×10^{25}	0.7199	$H^{0.6401}.v_e^{0.3599}$
75	Re	186.21	4.194×10^{25}	0.7200	$H^{0.6400}.v_e^{0.3600}$
76	Os	190.23	4.285×10^{25}	0.7201	$H^{0.6399}.v_e^{0.3601}$
77	Ir	192.22	4.329×10^{25}	0.7202	$H^{0.6399}.v_e^{0.3601}$
78	Pt	195.08	4.394×10^{25}	0.7203	$H^{0.6398}.v_e^{0.3602}$
79	Au	196.97	4.436×10^{25}	0.7204	$H^{0.6398}.v_e^{0.3602}$
80	Hg	200.59	4.518×10^{25}	0.7205	$H^{0.6397}.v_e^{0.3603}$
81	Tl	204.38	4.603×10^{25}	0.7207	$H^{0.6397}.v_e^{0.3603}$
82	Pb	207.2	4.667×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
83	Bi	208.98	4.707×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
84	Po	209	4.707×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
85	At	210	4.730×10^{25}	0.7209	$H^{0.6396}.v_e^{0.3604}$
86	Rn	222	5.000×10^{25}	0.7213	$H^{0.6394}.v_e^{0.3606}$
87	Fr	223	5.023×10^{25}	0.7213	$H^{0.6394}.v_e^{0.3606}$
88	Ra	226	5.090×10^{25}	0.7214	$H^{0.6393}.v_e^{0.3607}$
89	Ac	227	5.113×10^{25}	0.7214	$H^{0.6393}.v_e^{0.3607}$
90	Th	232.04	5.226×10^{25}	0.7216	$H^{0.6392}.v_e^{0.3608}$
91	Pa	231.04	5.204×10^{25}	0.7215	$H^{0.6392}.v_e^{0.3608}$
92	U	238.03	5.361×10^{25}	0.7218	$H^{0.6391}.v_e^{0.3609}$
93	Np	237	5.338×10^{25}	0.7217	$H^{0.6391}.v_e^{0.3609}$
94	Pu	244	5.496×10^{25}	0.7219	$H^{0.6390}.v_e^{0.3610}$
95	Am	243	5.473×10^{25}	0.7219	$H^{0.6390}.v_e^{0.3610}$
96	Cm	247	5.563×10^{25}	0.7220	$H^{0.6390}.v_e^{0.3610}$
97	Bk	247	5.563×10^{25}	0.7220	$H^{0.6390}.v_e^{0.3610}$
98	Cf	251	5.653×10^{25}	0.7221	$H^{0.6389}.v_e^{0.3611}$

99	Es	252	5.676×10^{25}	0.7222	$H^{0.6389}.\nu_e{}^{0.3611}$
100	Fm	257	5.789×10^{25}	0.7223	$H^{0.6388}.\nu_e{}^{0.3612}$
101	Md	258	5.811×10^{25}	0.7223	$H^{0.6388}.\nu_e{}^{0.3612}$
102	No	259	5.834×10^{25}	0.7224	$H^{0.6388}.\nu_e{}^{0.3612}$
103	Lr	266	5.991×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
104	Rf	267	6.014×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
105	Db	268	6.036×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
106	Sg	269	6.059×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
107	Bh	270	6.081×10^{25}	0.7227	$H^{0.6387}.\nu_e{}^{0.3613}$
108	Hs	269	6.059×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
109	Mt	278	6.262×10^{25}	0.7229	$H^{0.6386}.\nu_e{}^{0.3614}$
110	Ds	281	6.329×10^{25}	0.7230	$H^{0.6385}.\nu_e{}^{0.3615}$
111	Rg	282	6.352×10^{25}	0.7230	$H^{0.6385}.\nu_e{}^{0.3615}$
112	Cn	285	6.419×10^{25}	0.7231	$H^{0.6385}.\nu_e{}^{0.3615}$
113	Nh	286	6.442×10^{25}	0.7231	$H^{0.6385}.\nu_e{}^{0.3615}$
114	Fl	289	6.509×10^{25}	0.7232	$H^{0.6384}.\nu_e{}^{0.3616}$
115	Mc	290	6.532×10^{25}	0.7232	$H^{0.6384}.\nu_e{}^{0.3616}$
116	Lv	293	6.599×10^{25}	0.7233	$H^{0.6384}.\nu_e{}^{0.3616}$
117	Ts	294	6.622×10^{25}	0.7233	$H^{0.6384}.\nu_e{}^{0.3616}$
118	Og	294	6.622×10^{25}	0.7233	$H^{0.6384}.\nu_e{}^{0.3616}$

5. Status

Predictions P1–P5 and the disclosure of data already examined are stated in the preregistration text. Sections 0–5 add no new prediction; they record the formulas and present values as of 2026-11-05. Section 6 registers one additional prediction, P6, on 2026-11-06.

6. Addendum — prediction P6, the Planck gate (registered 2026-11-06; P1–P5 unchanged)

The Planck mass $m_P = \sqrt{(\hbar c/G)} = 21.8$ μg has total frequency m_Pc^2/h exactly equal to the human-boundary (Planck) frequency ν_P. Reading: an ether structure whose total frequency exceeds ν_P can no longer exist as a wave at the human rung.

Item	Fixed statement

Prediction	No mechanical degree of freedom with effective mass ≥ 21.8 μg can be prepared in a non-classical motional state.
Effective mass	The effective (motional) mass m_{eff} of the mode as reported by the experiment; for centre-of-mass motion, the total mass.
Non-classical state (either suffices)	(i) Wigner-function negativity of the motional state at ≥ 3σ; (ii) a superposition of two coherent states (cat state) whose components are separated in phase space by more than twice the zero-point amplitude, evidenced by interference fringes or Wigner negativity.
Does not count	Ground-state cooling, squeezing, Gaussian entanglement — Gaussian states with everywhere-positive Wigner functions (e.g. near-ground-state cooling of the 10 kg LIGO mirrors is not a counterexample).
Falsified if	A peer-reviewed experiment reports (i) or (ii) for a mechanical mode with $m_{eff} \geq 21.8$ μg.
Expected trend (not a falsification condition)	Approaching 21.8 μg, cat-state coherence times fall faster than environmental decoherence models predict, however good the isolation.
Deciding data	High-overtone bulk acoustic resonators coupled to superconducting qubits; levitated particles; massive optomechanics.
Data seen before fixing	Heaviest known: Bild et al. 2023 (Science 380, 274), cat states of a mode with $m_{eff} = 16.2$ μg, 74% of the gate; no heavier result found on 2026-11-06.
Risk	Standard quantum mechanics predicts no such gate, and no known obstacle prevents scaling the same technique beyond 22 μg; P6 may be tested within years and may well fail.

6.1 Addendum — prediction P6.1, the Planck blind band and observational collective effect (registered 2026-11-06; P1–P6 unchanged)

Proposition and physical mechanism (the observational collective): By Axioms 1, 3, and 4, an observation is never an isolated object acting alone: the target matter (near 21.8 μg), the probing light, the vacuum, the measuring apparatus, the observer's eye and body (whose body mass M_{obs} carries its own ether frequency $F = M_{obs}c^2/h$), and the surrounding laboratory structure and nearby heavy equipment are all distinct ether structures with their own frequencies and overlapping ether spaces. Human observation is the field-pressure coupling across this entire **observational collective**. When the effective mass of a target mode enters the neighbourhood of the human-boundary frequency $\nu_P = 2.952\times10^{42}$ Hz (corresponding to the Planck mass $m_P = 21.8$ μg), the collective frequency of the coupled system gates human-layer readout at ν_P: field-pressure transduction decouples, causing **both classical and non-classical observations to vanish simultaneously**. Because the exact decoupling threshold is set by the collective frequency of the entire coupled environment, the blind region is not a mathematical point at 21.764 μg but a **finite mass interval whose centre and width shift with the laboratory environment, nearby large-scale equipment, and the body-mass frequencies of present observers**.

Item	Fixed statement
Prediction (dual-channel blind band)	Within a finite mass window around $m_P = 21.8$ μg (**21.8 ± 0.3 μg, i.e. 21.5–22.1 μg** under baseline laboratory conditions, and wider under strong collective coupling), field-pressure coupling between the target mode and the human observational collective closes: **both classical observables (optical scattering, thermal emission, centre-of-mass readout) and non-classical observables (quantum interference, Wigner negativity) vanish**

	simultaneously. Outside this window (< 21.5 μg or > 22.1 μg), classical observability resumes while non-classical states remain absent above 22.1 μg (P6).
The observational collective (why it is an interval)	The target mass, probing light, vacuum, detector, observer's eye and body ($F_{body} = M_{body}c^2/h$), laboratory building, and surrounding heavy apparatus are all ether structures whose ether flows superpose in quadrature ($v^2 = 2\Phi$, Axiom 4) to form a **collective frequency field**. This collective coupling broadens the point threshold m_P into a finite band (± 0.3 μg or wider).
Collective environmental shift (sub-prediction)	Because the blind band is governed by the observational collective, repeating the measurement on the same target across **different laboratories, toggling or relocating nearby large-scale equipment (high-field magnets, heavy machinery, high-power RF sources), or varying the presence and body masses (frequencies) of near-field human observers** will produce a **reproducible shift in both the centre mass and the width of the blind window**.
Effective mass	The reported effective (motional) mass m_{eff} of the mode, or total mass for centre-of-mass / levitated-particle measurements.
Observational disappearance (both required)	(i) Inside the collective blind window, the effective scattering cross-section or readout signal across both classical and non-classical channels drops to **< 1% of its out-of-band baseline (or below instrument noise floor)**; (ii) tuning m_{eff} or testing control samples just outside the window restores normal classical scattering and mechanical readout (ruling out sample loss or damage).
Does not count	Signal loss caused by sample evaporation, mechanical detachment, optical misalignment, cavity detuning, or ordinary detector malfunction.
Falsified if	A peer-reviewed experiment scanning across **21.0–22.6 μg** (covering 21.8 ± 0.3 μg) observes smooth, continuous classical optical/mechanical scattering and readout throughout the entire interval without any anomalous dip (< 10% attenuation).
Expected trend (not a falsification condition)	Approaching the collective blind band from either side produces a sharp edge drop in both classical signal-to-noise ratio and non-classical coherence time; the window widens when heavy surrounding equipment is active or multiple observers are in close proximity compared to remote unmanned operation.
Deciding data	Continuously tunable bulk-acoustic / optomechanical resonators spanning 16–25 μg, microgram-scale optical/magnetic levitation scattering scans, and differential runs comparing **different laboratories, surrounding heavy-equipment states, and manned (varying observer body mass) vs. unmanned remote readout**.
Data seen & risk	Existing quantum optomechanics reaches 16.2 μg (Bild et al. 2023), while macroscopic micro-particle scattering typically probes outside 21.5–22.1 μg without fine mass-stepped coupling scans. Standard physics predicts completely smooth, environment- and observer-independent classical scattering across 21.8 μg; P6.1 can therefore be falsified within months by a straightforward microgram-sphere scattering scan.